

''' Name : Amit Kumar, Regd No. : 2341011129, Sl No. : 22 '''

Q1

Natural Language Processing (NLP) is a field of Artificial Intelligence that enables computers to understand, interpret, and generate human language. Applications:

1. Chatbots – Used in customer service to automate responses and improve user experience.
2. Language Translation – Tools like Google Translate help break language barriers globally.
3. Sentiment Analysis – Used in social media and marketing to analyze public opinion. Impact on society: NLP improves communication, automates tasks, and enhances accessibility of information.

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Q2

Tokenization: Breaking text into words or sentences. It is the first step in NLP. Stemming: Reducing words to their root form by removing suffixes. Lemmatization: Converting words to their meaningful base form using vocabulary and grammar rules. Significance: These processes help machines understand and normalize text for better analysis.

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Q3

Part-of-Speech (POS) tagging is the process of labeling words with their grammatical roles such as noun, verb, adjective, etc.

Importance: It helps in understanding sentence structure and meaning. Example: Sentence: "She eats apples" POS Tags: She(PRON), eats(VERB), apples(NOUN)

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```

```
# Q4
from textblob import TextBlob
exercise_blob = TextBlob("This is a TextBlob")
print(exercise_blob)
```

```
This is a TextBlob
```

```
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```

```
# Q5
import nltk
from nltk.tokenize import word_tokenize, sent_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer, WordNetLemmatizer

nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')

text = """Natural Language Processing enables machines to understand and process human languages.
It is a fascinating field with numerous applications, such as chatbots and language translation."""
words = word_tokenize(text)
sentences = sent_tokenize(text)
stop_words = set(stopwords.words("english"))
filtered_words = [w for w in words if w.lower() not in stop_words and w.isalpha()]

stemmer = PorterStemmer()
stemmed = [stemmer.stem(w) for w in filtered_words]
lemmatizer = WordNetLemmatizer()
lemmatized = [lemmatizer.lemmatize(w) for w in filtered_words]

print("Words:", words)
print("Sentences:", sentences)
print("After Stopword Removal:", filtered_words)
print("Stemmed:", stemmed)
print("Lemmatized:", lemmatized)
```

```
Words: ['Natural', 'Language', 'Processing', 'enables', 'machines', 'to', 'understand', 'and', 'process', 'human', 'language']
Sentences: ['Natural Language Processing enables machines to understand and process human languages.', 'It is a fascinating field with numerous applications, such as chatbots and language translation.']
After Stopword Removal: ['Natural', 'Language', 'Processing', 'enables', 'machines', 'understand', 'process', 'human', 'lang
```

```
Stemmed: ['natur', 'langaug', 'process', 'enabl', 'machin', 'understand', 'process', 'human', 'langaug', 'fascin', 'field',
Lemmatized: ['Natural', 'Language', 'Processing', 'enables', 'machine', 'understand', 'process', 'human', 'language', 'fasci
[nltk_data] Downloading package punkt to
[nltk_data] C:\Users\chinu\AppData\Roaming\nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data] C:\Users\chinu\AppData\Roaming\nltk_data...
[nltk_data] Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to
[nltk_data] C:\Users\chinu\AppData\Roaming\nltk_data...
[nltk_data] Package wordnet is already up-to-date!
```

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```
# Q6
import requests
from bs4 import BeautifulSoup
from nltk.corpus import stopwords
from wordcloud import WordCloud
import matplotlib.pyplot as plt
import nltk

nltk.download('stopwords')
url = "https://www.python.org"
response = requests.get(url)
soup = BeautifulSoup(response.text, "html.parser")
text = soup.get_text()
words = text.split()
stop_words = set(stopwords.words("english"))
filtered = [w for w in words if w.lower() not in stop_words]
clean_text = " ".join(filtered)
wc = WordCloud(width=800, height=400, background_color="white").generate(clean_text)
plt.imshow(wc, interpolation="bilinear")
plt.axis("off")
plt.show()
```

```
[nltk_data] Downloading package stopwords to
[nltk_data] C:\Users\chinu\AppData\Roaming\nltk_data...
[nltk_data] Package stopwords is already up-to-date!
```



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```

```
# Q7
import nltk
nltk.download('brown')
from textblob import TextBlob
```

```
text = "Natural Language Processing enables machines to understand human language. It is widely used in AI applications."
```

```
blob = TextBlob(text)
print(blob.sentences)
print(blob.words)
print(blob.noun_phrases)
```

```
[nltk_data] Downloading package brown to
[nltk_data] C:\Users\chinu\AppData\Roaming\nltk_data...
[nltk_data] Unzipping corpora\brown.zip.
[Sentence("Natural Language Processing enables machines to understand human language."), Sentence("It is widely used in AI a
['Natural', 'Language', 'Processing', 'enables', 'machines', 'to', 'understand', 'human', 'language', 'It', 'is', 'widely',
['language processing', 'human language', 'ai']
```

```
''' Name : Amit Kumar, Regd No. : 2341011129, Sl No. : 22 '''
```

```
# Q8
import requests
from bs4 import BeautifulSoup
from textblob import TextBlob
```

```
url = "https://www.bbc.com/news"
text = BeautifulSoup(requests.get(url).text, "html.parser").get_text()
```

```
blob = TextBlob(text)
print("Overall Sentiment:", blob.sentiment)
for s in blob.sentences[:5]:
    print(s.sentiment)
```

```
Overall Sentiment: Sentiment(polarity=0.06127609427609428, subjectivity=0.44983501683501687)
Sentiment(polarity=0.5, subjectivity=0.7)
Sentiment(polarity=0.0, subjectivity=0.0)
Sentiment(polarity=-0.0015151515151515188, subjectivity=0.5171717171717172)
Sentiment(polarity=-0.035311447811447814, subjectivity=0.3966750841750842)
Sentiment(polarity=0.25, subjectivity=0.3333333333333333)
```

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```
# Q9
import requests
from bs4 import BeautifulSoup
from textblob import TextBlob
from textblob.sentiments import NaiveBayesAnalyzer
import nltk
nltk.download('movie_reviews')
nltk.download('punkt')
url = "https://www.bbc.com/news"
text = BeautifulSoup(requests.get(url).text, "html.parser").get_text()
blob = TextBlob(text, analyzer=NaiveBayesAnalyzer())
print("Overall Sentiment:", blob.sentiment)
for s in blob.sentences[:5]:
    print(TextBlob(str(s), analyzer=NaiveBayesAnalyzer()).sentiment)
```

```
[nltk_data] Downloading package movie_reviews to
[nltk_data] C:\Users\chinu\AppData\Roaming\nltk_data...
[nltk_data] Package movie_reviews is already up-to-date!
[nltk_data] Downloading package punkt to
[nltk_data] C:\Users\chinu\AppData\Roaming\nltk_data...
[nltk_data] Package punkt is already up-to-date!
Overall Sentiment: Sentiment(classification='pos', p_pos=1.0, p_neg=4.484184267816143e-25)
Sentiment(classification='pos', p_pos=0.6569575857307592, p_neg=0.3430424142692426)
Sentiment(classification='pos', p_pos=0.5, p_neg=0.5)
Sentiment(classification='pos', p_pos=0.9998396441885559, p_neg=0.0001603558114608325)
Sentiment(classification='pos', p_pos=0.9999998802906337, p_neg=1.197093627341511e-07)
Sentiment(classification='pos', p_pos=0.7901658397031855, p_neg=0.2098341602968183)
```

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```
# Q10
import requests
from textblob import TextBlob

url = "https://www.gutenberg.org/files/1342/1342-0.txt"
text = requests.get(url).text[:5000]

blob = TextBlob(text)

words = blob.words

for w in words[:100]:
    if w.lower() != w.correct():
        print(w, "->", w.spellcheck()[0][3])
```

```
START -> [('START', 0.0)]
THE -> [('THE', 0.0)]
PROJECT -> [('PROJECT', 0.0)]
GUTENBERG -> [('GUTENBERG', 0.0)]
EBOOK -> [('EBOOK', 0.0)]
Illustration -> [('Illustration', 1.0)]
GEORGE -> [('GEORGE', 0.0)]
ALLEN -> [('ALLEN', 0.0)]
PUBLISHER -> [('PUBLISHER', 0.0)]
CHARING -> [('CHARING', 0.0)]
CROSS -> [('CROSS', 0.0)]
ROAD -> [('ROAD', 0.0)]
LONDON -> [('LONDON', 0.0)]
RUSKIN -> [('RUSKIN', 0.0)]
HOUSE -> [('HOUSE', 0.0)]
Illustration -> [('Illustration', 1.0)]
Reading -> [('Leading', 0.4791666666666667), ('Reading', 0.475), ('Heading', 0.04166666666666667)]
Jane -> [('Lane', 0.5625), ('Mane', 0.1875), ('Cane', 0.10416666666666667)]
Letters -> [('Letters', 0.9908256880733946), ('Fetters', 0.009174311926605505)]
Chap -> [('Chap', 1.0)]
PRIDE -> [('PRIDE', 0.0)]
PREJUDICE -> [('PREJUDICE', 0.0)]
Jane -> [('Lane', 0.5625), ('Mane', 0.1875), ('Cane', 0.10416666666666667)]
```

```

Austen -> [('Listen', 0.704225352112676), ('Mustn', 0.09859154929577464), ('Hasten', 0.07042253521126761)]
Preface -> [('Preface', 1.0)]
George -> [('George', 1.0)]
Saintsbury -> [('Saintsbury', 0.0)]
Illustrations -> [('Illustrations', 1.0)]
Hugh -> [('Hugh', 0.7), ('Ugh', 0.3)]
Thomson -> [('Thomson', 1.0)]
Illustration -> [('Illustration', 1.0)]
Ruskin -> [('Skin', 0.8746223564954683), ('Tushin', 0.11329305135951662), ('Muslin', 0.0075528700906344415)]
Charing -> [('Sharing', 0.92), ('Charing', 0.08)]
House -> [('House', 0.977810650887574), ('Rouse', 0.011834319526627219), ('Mouse', 0.008875739644970414)]
Cross -> [('Cross', 0.6956521739130435), ('Gross', 0.2028985507246377), ('Ross', 0.10144927536231885)]
Road -> [('Road', 0.9701492537313433), ('Load', 0.029850746268656716)]
London -> [('London', 1.0)]
George -> [('George', 1.0)]
Allen -> [('Ellen', 0.5), ('Allen', 0.5)]
CHISWICK -> [('CHISWICK', 0.0)]
PRESS -> [('PRESS', 0.0)]
CHARLES -> [('CHARLES', 0.0)]
WHITTINGHAM -> [('WHITTINGHAM', 0.0)]
AND -> [('AND', 0.0)]
CO -> [('of', 0.17512732555086896), ('to', 0.12586415107548524), ('in', 0.0964786390605037)]
TOOKS -> [('TOOKS', 0.0)]
COURT -> [('COURT', 0.0)]
CHANCERY -> [('CHANCERY', 0.0)]
LANE -> [('LANE', 0.0)]
LONDON -> [('LONDON', 0.0)]
Illustration -> [('Illustration', 1.0)]
To -> [('To', 0.7805182471849138), ('So', 0.0818613485280152), ('No', 0.06370913037579705)]
J -> [('J', 1.0)]
Comyns -> [('Romans', 0.75), ('Omens', 0.25)]
Carr -> [('Parr', 1.0)]
I -> [('I', 1.0)]
Hugh -> [('Hugh', 0.7), ('Ugh', 0.3)]
Thomson -> [('Thomson', 1.0)]

```

```

''' Name : Amit Kumar, Regd No. : 2341011129, Sl No. : 22 '''

# Q11
from textblob import TextBlob, Word

# A) (replaced translate) Basic sentiment of a predefined text
text = "Natural Language Processing is amazing"
blob = TextBlob(text)
print("Polarity:", blob.sentiment.polarity)
print("Subjectivity:", blob.sentiment.subjectivity)

# B) Sentiment classification
sentences = [
    "I love AI",
    "This is a book",
    "I hate bugs"
]
for text in sentences:
    blob = TextBlob(text)
    polarity = blob.sentiment.polarity
    subjectivity = blob.sentiment.subjectivity
    obj = "Objective" if subjectivity < 0.5 else "Subjective"
    if polarity > 0:
        senti = "Positive"
    elif polarity < 0:
        senti = "Negative"
    else:
        senti = "Neutral"
    print(obj, senti)

# C) Sentence-wise sentiment
text = "NLP is interesting. It is widely used."
blob = TextBlob(text)
for sentence in blob.sentences:
    print(sentence)
    print(sentence.sentiment)

# D) POS tagging
text = "TextBlob makes NLP easy"
blob = TextBlob(text)
for word, tag in blob.tags:
    print(word, tag)

# E) Spell check
word = "speling"
w = Word(word)
print(w.spellcheck())
print(w.correct())

# F) Extract adjectives

```

```

text = "This is a beautiful and amazing day"
blob = TextBlob(text)
adjectives = [word for word, tag in blob.tags if tag == "JJ"]
print(adjectives)

# G) Top noun phrases
text = "Natural language processing is a field of artificial intelligence. Natural language tools are useful."
blob = TextBlob(text)
freq = {}
for p in blob.noun_phrases:
    freq[p] = freq.get(p, 0) + 1

top5 = sorted(freq.items(), key=lambda x: x[1], reverse=True)[:5]
print(top5)

# H) Simple summarization
text = "Natural language processing is powerful. It is used in chatbots. Chatbots are useful tools."
blob = TextBlob(text)
freq = {}
for p in blob.noun_phrases:
    freq[p] = freq.get(p, 0) + 1
important = set([p[0] for p in sorted(freq.items(), key=lambda x: x[1], reverse=True)[:3]])
summary = []
for sentence in blob.sentences:
    for p in important:
        if p in sentence.lower():
            summary.append(str(sentence))
            break
print(summary)

```

```

Polarity: 0.35000000000000003
Subjectivity: 0.65
Subjective Positive
Objective Neutral
Subjective Negative
NLP is interesting.
Sentiment(polarity=0.5, subjectivity=0.5)
It is widely used.
Sentiment(polarity=-0.1, subjectivity=0.4)
TextBlob NNP
makes VBZ
NLP NNP
easy JJ
[('spelling', 1.0)]
spelling
['beautiful', 'amazing']
[('natural language processing', 1), ('artificial intelligence', 1), ('natural language tools', 1)]
['Natural language processing is powerful.', 'It is used in chatbots.', 'Chatbots are useful tools.']

```

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```

```

# Q12
from textblob import Word
w = Word("happy")
print("Definition:", w.definitions)
syn = []
ant = []
for s in w.synsets:
    for l in s.lemmas():
        syn.append(l.name())
        if l.antonyms():
            ant.append(l.antonyms()[0].name())
print("Synonyms:", set(syn))
print("Antonyms:", set(ant))

```

```

Definition: ['enjoying or showing or marked by joy or pleasure', 'marked by good fortune', 'eagerly disposed to act or to be
Synonyms: {'glad', 'well-chosen', 'happy', 'felicitous'}
Antonyms: {'unhappy'}

```

```
''' Name : Amit Kumar, Regd No. : 2341011129, Sl No. : 22 '''
```

```

# Q13A
from wordcloud import WordCloud
import matplotlib.pyplot as plt
text = open("input.txt", "r", encoding="utf-8").read()
wc = WordCloud(width=800, height=400, background_color="white").generate(text)
plt.imshow(wc)
plt.axis("off")
plt.show()

```

used interesting AI Language Natural Processing

''' Name : Amit Kumar, Regd No. : 2341011129, Sl No. : 22 '''

```
# Q13B
from wordcloud import WordCloud
import matplotlib.pyplot as plt
import numpy as np
from PIL import Image
text = open("input.txt", "r", encoding="utf-8").read()
mask = np.array(Image.open("mask.png"))
wc = WordCloud(background_color="white", mask=mask).generate(text)
plt.imshow(wc)
plt.axis("off")
plt.show()
```

Processing

used
Natural
Language
AI
interesting

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```
# Q14
import requests
from bs4 import BeautifulSoup
import nltk
nltk.download('punkt')
from nltk.tokenize import sent_tokenize, word_tokenize
urls = [
    "https://www.bbc.com/news",
    "https://www.cnn.com"
]
for url in urls:
    text = BeautifulSoup(requests.get(url).text, "html.parser").get_text()
    sentences = sent_tokenize(text)
    words = word_tokenize(text)
    avg_wps = len(words) / len(sentences)
    avg_cpw = sum(len(w) for w in words) / len(words)
    avg_spw = avg_cpw / 3
    print("\nURL:", url)
    print("Avg Words/Sentence:", avg_wps)
    print("Avg Chars/Word:", avg_cpw)
    print("Avg Syllables/Word:", avg_spw)
```

```
[nltk_data] Downloading package punkt to
[nltk_data] C:\Users\chinu\AppData\Roaming\nltk_data...
[nltk_data] Package punkt is already up-to-date!
```

```
URL: https://www.bbc.com/news
Avg Words/Sentence: 98.5
Avg Chars/Word: 5.9238578680203045
```

Avg Syllables/Word: 1.9746192893401016

URL: <https://www.cnn.com>

Avg Words/Sentence: 95.52173913043478

Avg Chars/Word: 4.875739644970414

Avg Syllables/Word: 1.6252465483234715

''' Name : Amit Kumar, Regd No. : 2341011129, Sl No. : 22 '''

Q15

```
import requests
from bs4 import BeautifulSoup
import spacy
nlp = spacy.load("en_core_web_sm")
url = "https://www.bbc.com/news"
text = BeautifulSoup(requests.get(url).text, "html.parser").get_text()
doc = nlp(text)
for ent in doc.ents[:20]:
    print(ent.text, "-", ent.label_)
```

BBC News - Breaking - ORG
U.S. - GPE
BusinessTechnology of BusinessNYSE Opening BellTechnologyArtificial IntelligenceIntelligence RevolutionAI - ORG
MindTech - ORG
& TVMusicArt & DesignStyleBooksEntertainment NewsArtsArts - ORG
MotionTravelDestinationsAfricaAntarcticaAsiaAustralia - GPE
PacificCaribbean & BermudaCentral AmericaEuropeMiddle EastNorth AmericaSouth AmericaWorld's TableCulture & - ORG
SportHomeNewsSportBusinessTechnologyHealthCultureArtsTravelEarthAudioVideoLiveWeatherNewslettersNewsA decade - DATE
Trump - PERSON
ChinaBeijing - ORG
US - GPE
one - CARDINAL
says.8 hrs - PERSON
Wes Streeting - PERSON
Labour - PERSON
four - CARDINAL
Tuesday - DATE
Elon Musk - PERSON
Tim Cook - PERSON
Trump - ORG

''' Name : Amit Kumar, Regd No. : 2341011129, Sl No. : 22 '''

Q16

```
import requests
import spacy

nlp = spacy.load("en_core_web_md")
romeo = requests.get("https://www.gutenberg.org/files/1513/1513-0.txt").text[:50000]
comedy = requests.get("https://www.gutenberg.org/files/1526/1526-0.txt").text[:50000]
doc1 = nlp(romeo)
doc2 = nlp(comedy)
print("Similarity:", doc1.similarity(doc2))
```

Similarity: 0.9979163408279419

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Q17

```
from textblob.utils import strip_punc, lowerstrip
import requests
text = requests.get("https://www.gutenberg.org/files/1513/1513-0.txt").text
print(strip_punc(text, all=True)[:500])
print(lowerstrip(text, all=True)[:500])
```

START OF THE PROJECT GUTENBERG EBOOK 1513

THE TRAGEDY OF ROMEO AND JULIET

by William Shakespeare

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ACT III

Scene I A public Place

Scene I

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the tragedy of romeo and juliet

by william shakespeare

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